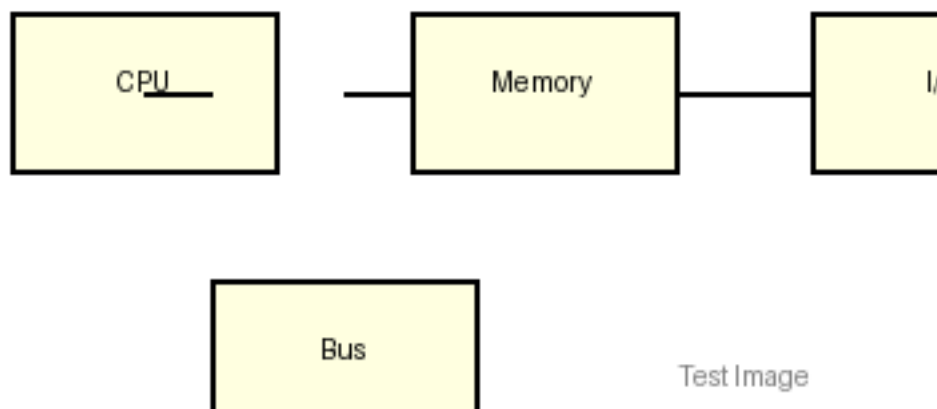




```
# Example: Initialize OCR Flow
from ocr_flow import Pipeline, Config

config = Config.load()
pipeline = Pipeline(config, verbose=True)

result = pipeline.run(
    input_pdf="datasheet.pdf",
    output_dir="./output",
    pdf_type="auto",
    language="en",
    translate=True
)
print(f"Output: {result}")
```

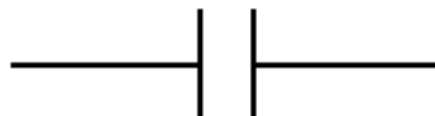
The OCR Flow system is built around a pipeline architecture. Each processing step is an independent module that can be enabled or disabled based on requirements.

Processing Steps:

1. PDF Type Detection - Determines if the PDF contains extractable text
2. OCR Processing - Uses UMI OCR for scanned documents
3. Translation - Optional BabelDOC integration for multilingual support
4. PDF Splitting - Divides large documents into manageable chunks
5. Compression - Reduces file size with Ghostscript
6. Markdown Conversion - Extracts structured content via MinerU API
7. Format Fixing - Applies formatting corrections
8. Image Download - Localizes remote images

State Management:

The pipeline maintains state in .state.json files, allowing for resume/retry functionality in case of failures.

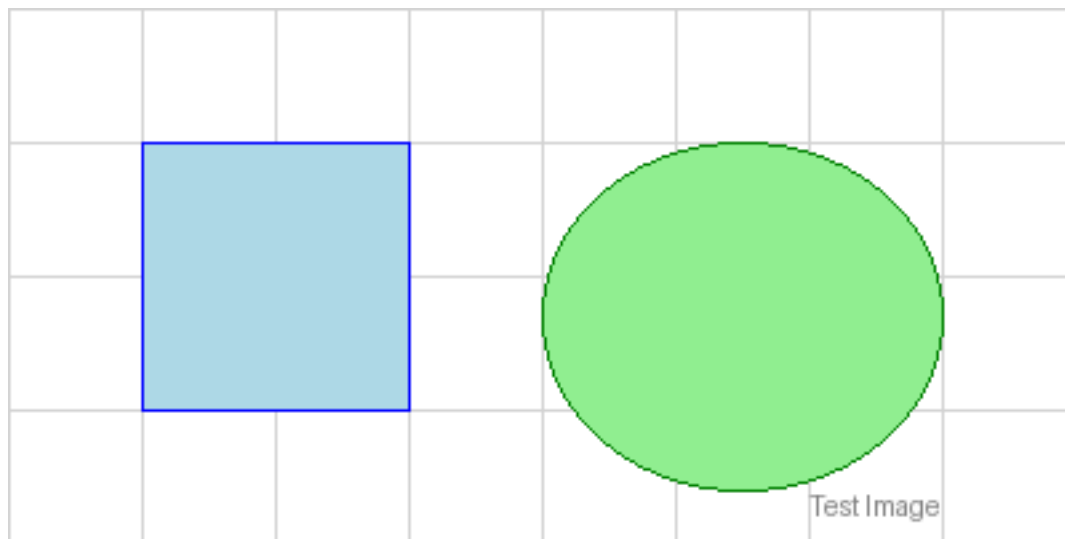


Test Image

```
# Pipeline state management
class State:
    def __init__(self, source_path: Path, options: Dict):
        self.source_path = str(source_path)
        self.options = options
        self.steps: Dict[str, StepStatus] = {}

    def update_step(self, step_name: str, status: str):
        self.steps[step_name] = StepStatus(status=status)
        self.save()

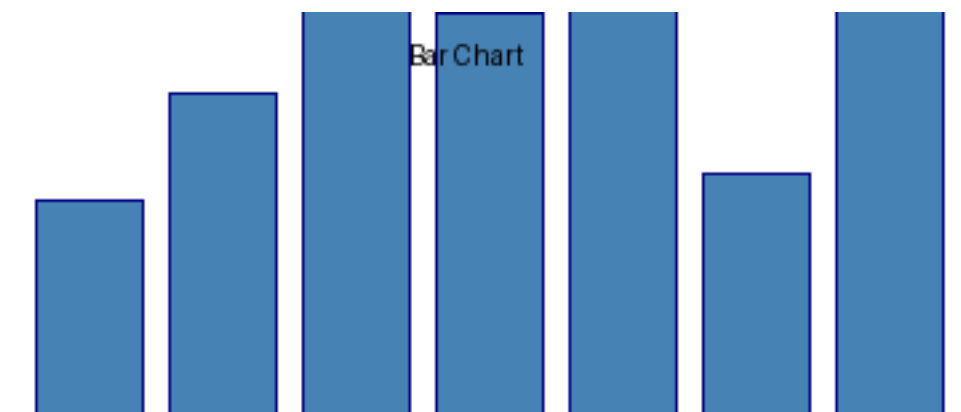
    def is_completed(self) -> bool:
        return all(s.status in ('completed', 'skipped')
                   for s in self.steps.values())
```



```
# config.toml example
[babeldoc]
path = "../BabelDOC"
lang_in = "en-US"
lang_out = "zh-CN"
openai = true
openai_model = "qwen3.5-flash"
openai_base_url = "https://dashscope.aliyuncs.com/compatible-mode/v1"
openai_api_key = "sk-xxx"

[umiocr]
url = "http://127.0.0.1:1224"
language = "models/config_en.txt"

[mineru]
api_token = "your-token-here"
```



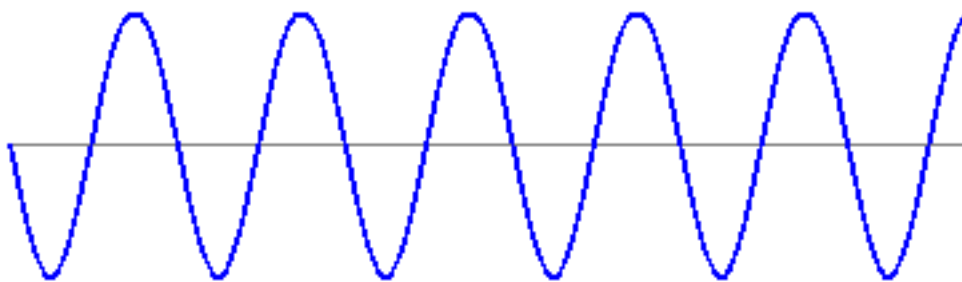
Test Image

```
// Memory map access example
#define GPIO_BASE      0x40000000
#define UART_BASE      0x40010000
#define SPI_BASE       0x40020000
#define I2C_BASE       0x40030000

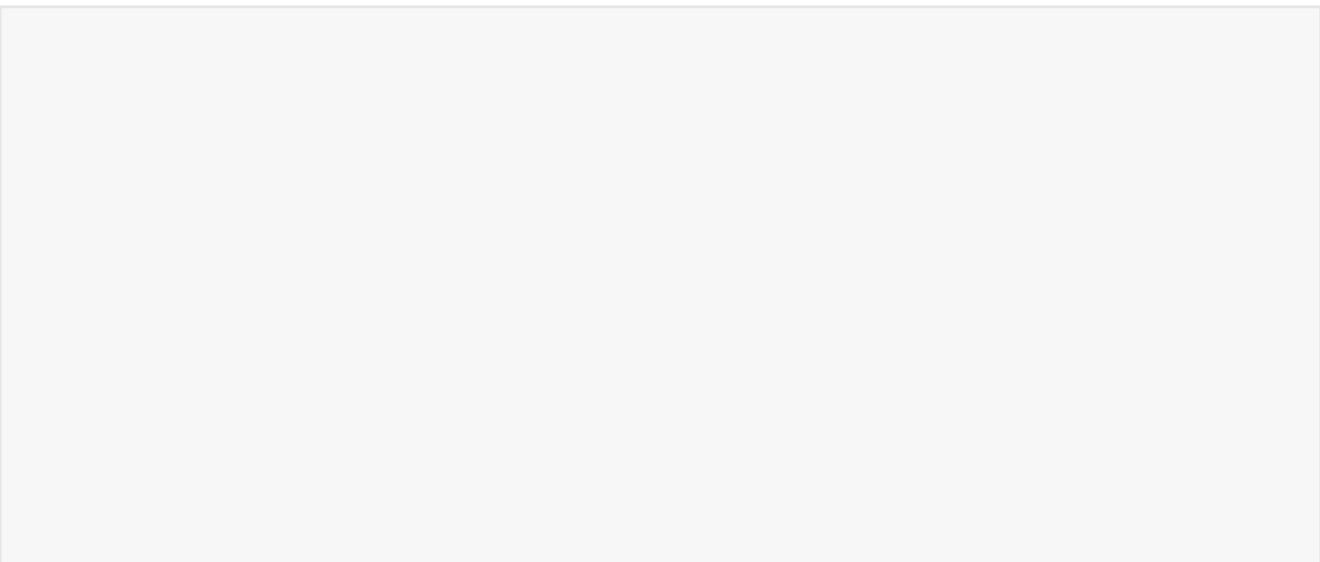
typedef struct {
    volatile uint32_t MODER;
    volatile uint32_t OTYPER;
    volatile uint32_t OSPEEDR;
    volatile uint32_t PUPDR;
    volatile uint32_t IDR;
    volatile uint32_t ODR;
} GPIO_TypeDef;

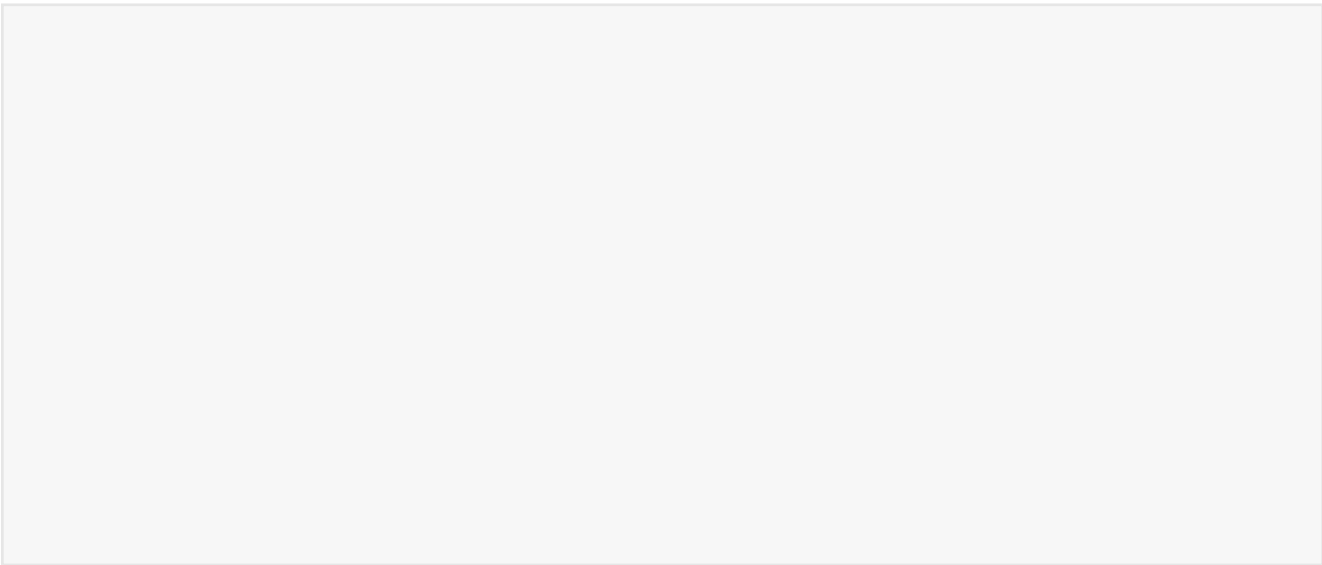
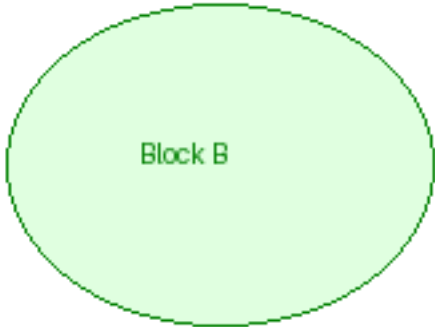
#define GPIOA ((GPIO_TypeDef*)GPIO_BASE)
```

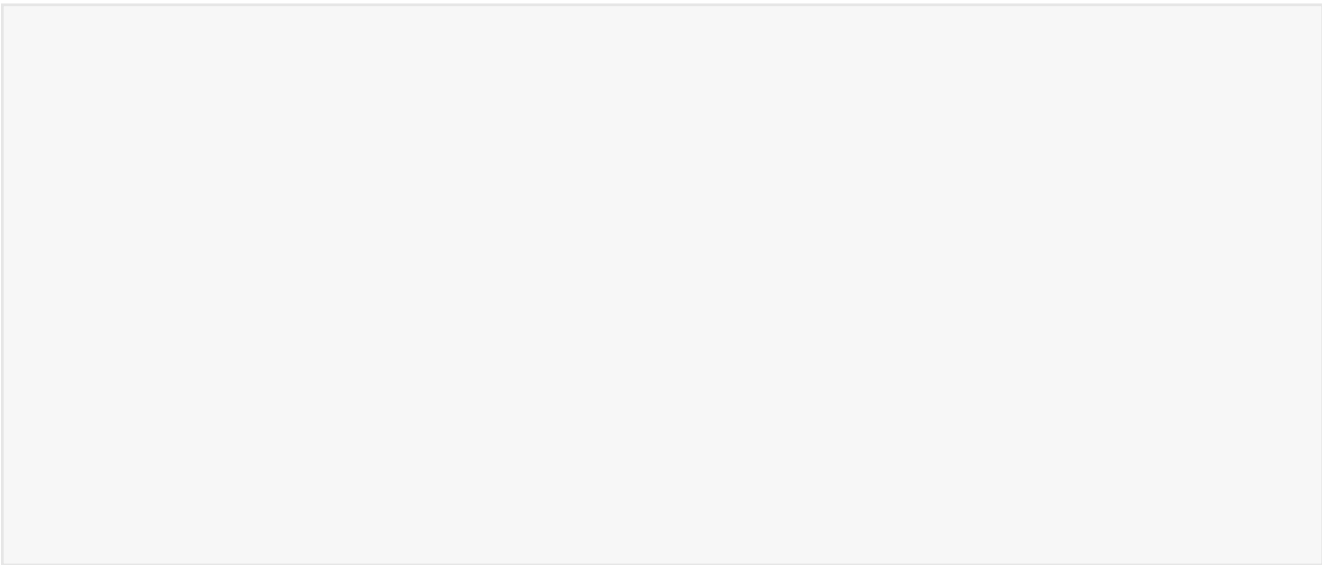
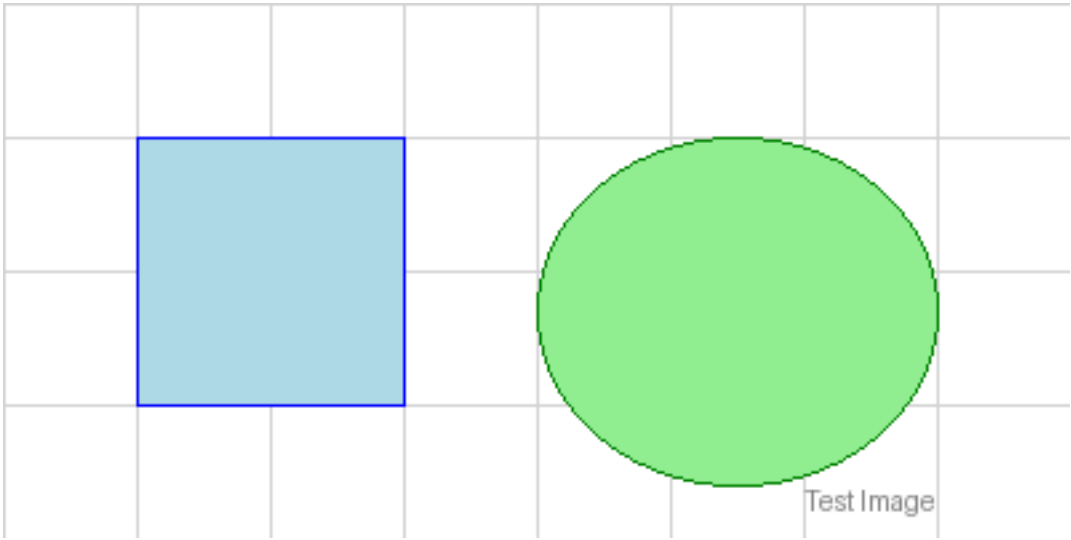
Sine Wave



Test Image

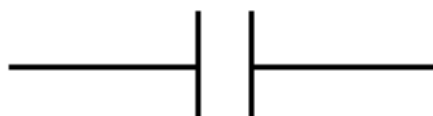






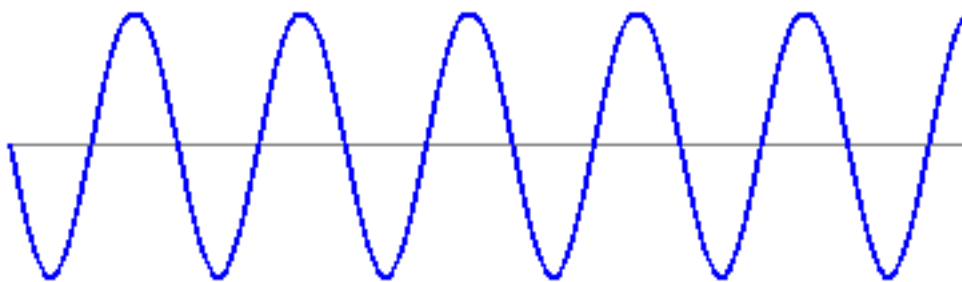


R1 = 10k Ω



Test Image

Sine Wave



Test Image

